



SLTC REASEARCH UNIVERSITY

MID – SEMESTER ASSIGNMENT SEMESTER 05

Bachelor of Applied Information Technology – (BAIT)

CIT308 – Data Visualization

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1.0 Business Questions and Intended Users

Lanka Mart is a multi-category retailer with a presence in 9 Provinces in Sri Lanka with 3 business models – stores, website and mobile app. Which provinces, product categories and channels bring the highest revenues and profits? (1) (2) Is there a consistency in profitability by level of discount and by segment of customers? (3) In which areas are returns made and what does this imply on product or fulfilment quality? The intended users are category managers who have to make a decision on which product lines to promote or de-prioritise and regional operations managers, who have to be able to see the performance and delivery/return patterns at province level to plan their logistics and after-sales processes.

2.0 Data Quality Checks and Transformations

The source file (CIT308_LankaMart_Retail_Transactions.csv) contains 721 order-line records for the period of January to June 2026 and with 19 fields. Prior to the construction of the dashboard, the data set was programmatically reviewed and cleansed as follows:

- There were 720 unique order lines after using the duplicated() function to remove one fully duplicated row (order_id and all attributes are identical) from pandas.
- Some inconsistent labelling of categories: one product category was marked as “electronic” (lower case, singular) while 149 were marked as “Electronics” (capital letter, plural). This was standardised to the canonical label so that category-level aggregates are not artificially split.
- It appears that 42 out of 720 rows (about 5.8%) had no customer_rating value. These were not imputed as zero or the mean for the column, as this would have skewed the average rating; they were left as missing and were not included in the calculations of the ratings or the rating-distribution chart, and so reported averages are based only on those customers who actually rated their order.
- No value in promotion field: 276 rows were missing promotion values. These were ignored on inspection, and not missing data or an unknown value, as they were only put in with the clearly marked label “None”.
- Consistency checks: The total revenue (revenue_lkr) and profit (profit_lkr) were verified with the units multiplied by unit_price_lkr multiplied by (1-discount_pct) and revenue_lkr minus cost_lkr respectively, and no arithmetic inconsistencies were detected; these were therefore used as provided.

Profit_margin (profit_lkr / revenue_lkr) and a delivery_band (where delivery_days was categorised into Same day, 1-3 days, and 4+ days) were calculated to help the analysis, because 234 out of 720 lines of orders had zero delivery days (store pickup or same day fulfilment).

3.0 Dashboard Design Rationale

The dashboard has been created in Python with Streamlit for layout and interactive Ness and Portly for charting; it is run locally, as required by the assignment brief. At the top of the KPI cards are four KPI cards - Total Revenue, Total Profit, Profit Margin, Return Rate - to help managers to see the headline performance before diving into the detail. Five chart types were selected under the KPIs to answer different questions: a monthly line chart to show the trend of revenue over time; a horizontal bar chart to show the revenue by product category (which was selected over a pie chart because there were five categories with similar values and could be easily sorted as horizontal bars); a horizontal bar chart to show the revenue by province; a bubble scatter plot to show the discount percentage against the profit margin, with the size of the bubble indicating the revenue, and the color indicating the product category; and a histogram of customer ratings. Every chart has the same single palette of the blue color, as the contrast and reduced eye strain with dark background, and all numbers are labelled in LKR or percentage format to avoid any ambiguity. Three sidebar filter options – order date range, province, sales channel – are all on one filtered data frame, and the detail table is updated at the same time, as though each filter is acting independently of the others – everything updates simultaneously when any filter changes; a “Reset all filters” button resets the view to its original, unfiltered state. The underlying filtered order lines are displayed in a detail table below the charts, and a written insights panel translates the filter state of the chart into plain language insights.

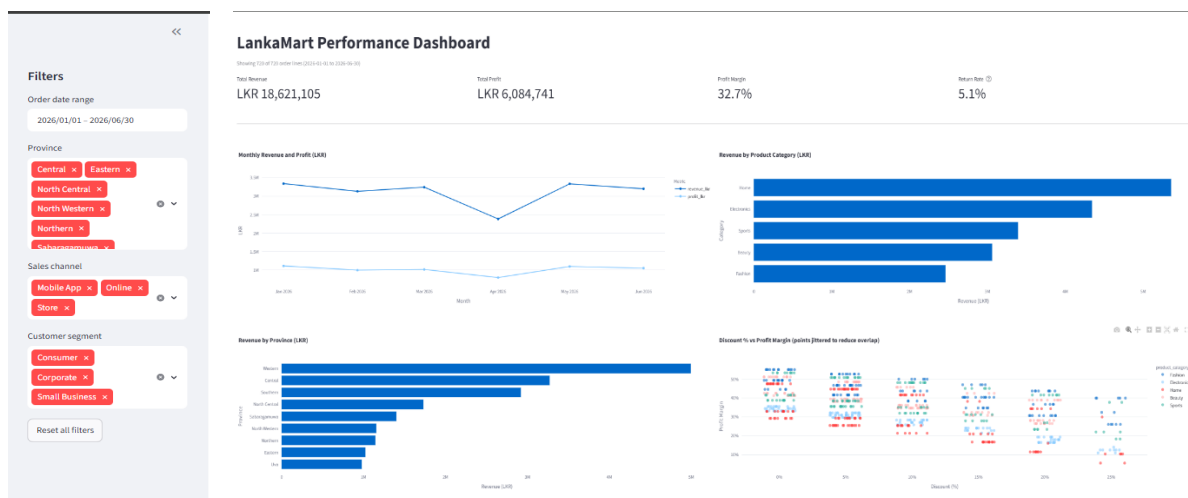




Figure 1: Dashboard, unfiltered baseline state (all 720 order lines Jan–Jun 26).

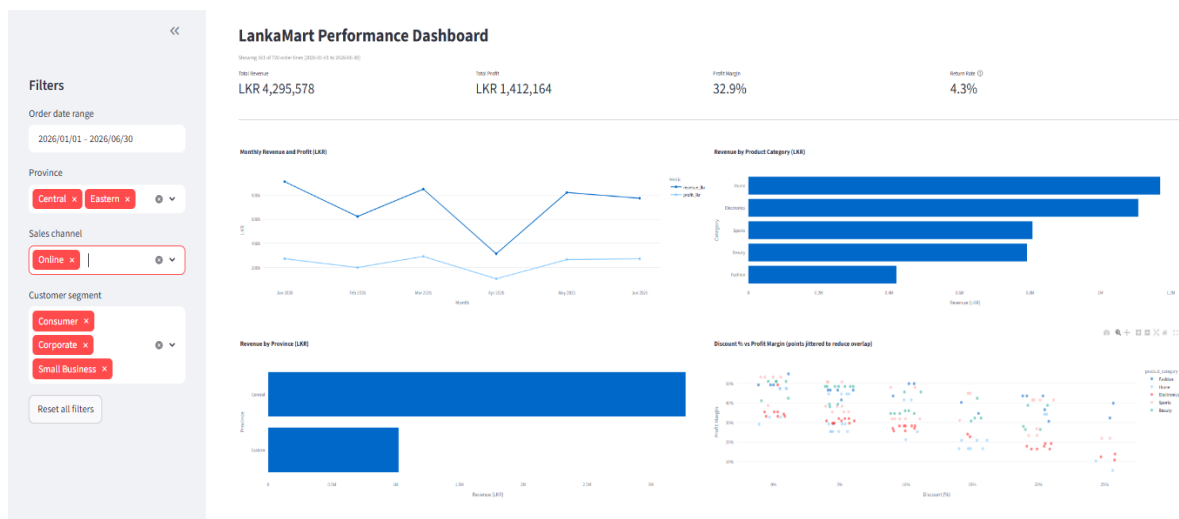


Figure 2 shows filtered state – Central and Eastern provinces, all sales channels (161 order lines).

4.0 Finding

For the entire unfiltered data set (720 order lines):

Total Revenue	LKR 18,621,105
Total Profit	LKR 6,084,741
Profit Margin	32.7%
Return Rate	5.1%

Finding 1 – Revenue is largely dominated by Western province and Home. The highest contribution to revenue comes from the western province with LKR 4,994,615 or about 27% of all the provinces' revenue. The category level, Home (LKR 5,360,392) is slightly ahead of Electronics (just behind), followed by Sports, Beauty and Fashion. Concentration would have

a disproportionate effect on revenue should the operations in the western provinces be impacted by a disruption, or the supply for the home category be impacted.

Finding 2 – Electronics has an unequal return. Electronics is the highest return rate of all product categories, at 10.0%, compared to the overall platform return rate of 5.1%. The platforms' top two revenue categories, Electronics and Online Gambling, also contribute to the top margin erosion risk type on the platform, making them a combined margin erosion risk.

Finding 3 – Pattern holds at regional level confirmed by filtering. In Central and Eastern provinces, all channels, the revenue is LKR 4,295,578, profit is LKR 1,412,164, profit margin is 32.9% and return is 4.3% which is close to the average achieved by the platform, suggesting that these two provinces are not outliers when it comes to profitability. Further filtering down to online orders only in Central and Eastern brings in revenue of LKR 1,557,795 (margin 32.6%, rate 5.7%) indicating that, on the whole, the online channel in these provinces is returning at a slightly higher rate than the province average, consistent with the general trend that non-physical channels return at a higher rate.

Finding 4 - Profit margin relatively flat by discount level the scatter plot of profit margin vs discount bands reveals that all the discount bands from 0% to 25% have profit margins that are grouped together in ranges between approximately 20% and 50%. This indicates that Lanka Mart's current discounting strategy is not that aggressive at this time, but outcomes spread further as discounts become more aggressive.

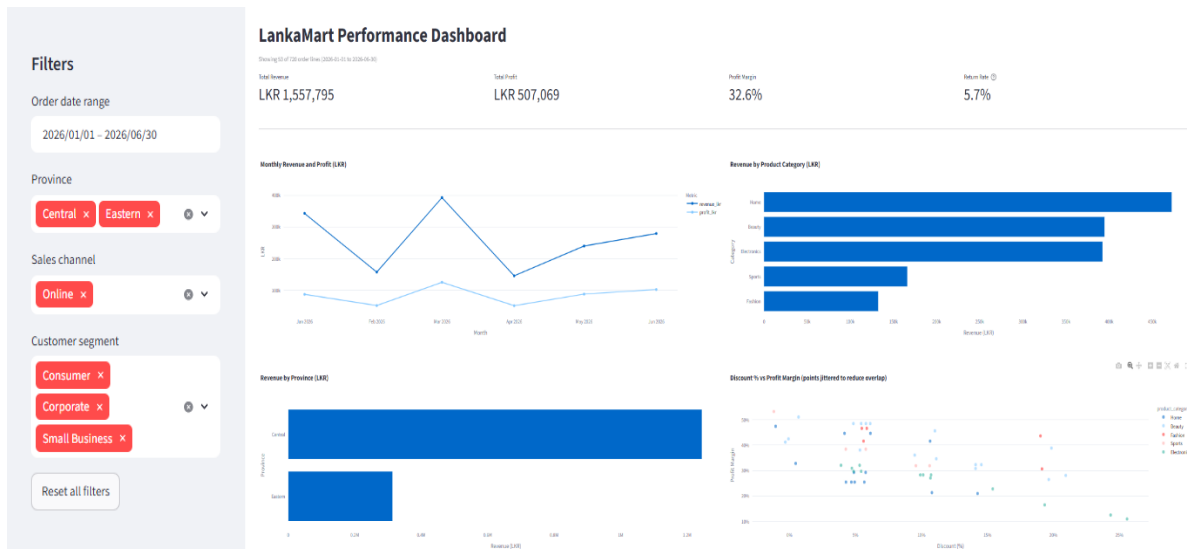


Figure 3: Filtered state – Central and Eastern provinces, online channel only (53 order lines).

order_id	order_date	province	city	sales_channel	customer_segment	product_category	product_name	units	revenue_bz	profit_bz	profit_margin	returned	delivery_days	customer_rating	promotion
LNKH-0003	2020-04-20 00:00:00	Central	Malabe	Online	Consumer	Home	Table Lamp	1	1090	2241	0.4127	No	41 days	5	App Exclusive
LNKH-0004	2020-04-20 00:00:00	Central	Kandy	Online	Consumer	Beauty	Skincare Set	3	37136	8176	0.1882	No	41 days	3	Weekend Deal
LNKH-0008	2020-04-21 00:00:00	Central	Malabe	Online	Consumer	Fashion	Backpack	4	23840	12042	0.3402	No	41 days	3	New Year
LNKH-0009	2020-04-20 00:00:00	Central	Kandy	Online	Consumer	Home	Rice Cooker	2	27088	9188	0.1251	No	41 days	2	None
LNKH-0019	2020-04-20 00:00:00	Central	Kandy	Online	Consumer	Fashion	Casual Shirt	4	12048	6048	0.4984	No	1-2 days	100	Weekend Deal
LNKH-0023	2020-04-22 00:00:00	Central	Nuwara Eliya	Online	Consumer	Home	Table Lamp	1	3035	2541	0.4402	No	1-2 days	4	App Exclusive
LNKH-0111	2020-04-21 00:00:00	Eastern	Dambulla	Online	Corporate	Sports	Dumbbell Set	4	37044	18044	0.3819	No	41 days	4	None
LNKH-0114	2020-04-20 00:00:00	Central	Kandy	Online	Small Business	Beauty	Skincare Set	3	17068	3608	0.1116	No	41 days	5	None
LNKH-0137	2020-04-20 00:00:00	Central	Kandy	Online	Consumer	Beauty	Hair Dryer	4	37048	8368	0.3089	No	1-2 days	5	Weekend Deal
LNKH-0138	2020-04-21 00:00:00	Central	Kandy	Online	Consumer	Home	Rice Cooker	3	24844	2744	0.1077	No	41 days	2	None
LNKH-0144	2020-04-20 00:00:00	Central	Malabe	Online	Small Business	Home	Air Fryer	2	47660	1260	0.0244	No	41 days	4	None
LNKH-0147	2020-04-20 00:00:00	Central	Kandy	Online	Consumer	Beauty	Skincare Set	3	24213	18213	0.4552	No	41 days	3	Weekend Deal
LNKH-0201	2020-04-20 00:00:00	Eastern	Dambulla	Online	Consumer	Beauty	Hair Dryer	1	7235	2851	0.1888	No	41 days	3	None
LNKH-0179	2020-04-21 00:00:00	Central	Malabe	Online	Consumer	Home	Rice Cooker	2	20550	7751	0.2787	No	41 days	4	Weekend Deal
LNKH-0193	2020-04-20 00:00:00	Central	Nuwara Eliya	Online	Consumer	Electronics	Smart Watch	4	71382	18416	0.2597	No	41 days	4	None
LNKH-0195	2020-04-20 00:00:00	Central	Nuwara Eliya	Online	Corporate	Electronics	Wireless Earbuds	1	9110	1210	0.1210	No	41 days	4	Weekend Deal
LNKH-0200	2020-04-20 00:00:00	Eastern	Dambulla	Online	Consumer	Electronics	Bluetooth Speaker	3	24910	4024	0.1214	No	1-2 days	5	App Exclusive
LNKH-0202	2020-04-20 00:00:00	Central	Nuwara Eliya	Online	Consumer	Home	Table Lamp	4	12668	1340	0.1058	No	41 days	4	None
LNKH-0206	2020-04-21 00:00:00	Central	Nuwara Eliya	Online	Corporate	Sports	Cycling Bag	4	40382	12982	0.1881	No	41 days	4	None
LNKH-0202	2020-04-20 00:00:00	Central	Nuwara Eliya	Online	Consumer	Electronics	Bluetooth Speaker	4	37762	13276	0.1338	No	1-2 days	4	None
LNKH-0209	2020-04-21 00:00:00	Eastern	Dambulla	Online	Consumer	Fashion	Running Shoes	3	18222	13022	0.4127	No	41 days	3	App Exclusive
LNKH-0203	2020-04-20 00:00:00	Central	Kandy	Online	Corporate	Beauty	Hair Dryer	3	12794	3384	0.2647	No	41 days	3	Weekend Deal
LNKH-0209	2020-04-20 00:00:00	Eastern	Dambulla	Online	Corporate	Beauty	Hair Dryer	2	14382	4882	0.3404	No	41 days	4	None
LNKH-0401	2020-04-20 00:00:00	Central	Malabe	Online	Consumer	Beauty	Perfume	4	20844	17018	0.1287	No	41 days	4	New Year
LNKH-0102	2020-04-20 00:00:00	Eastern	Dambulla	Online	Small Business	Home	Air Fryer	2	47660	1260	0.0244	No	1-2 days	4	None
LNKH-0103	2020-04-20 00:00:00	Eastern	Trincomalee	Online	Sports	Cycling Bag	1	12340	4141	0.1888	No	41 days	5	None	
LNKH-0105	2020-04-20 00:00:00	Central	Kandy	Online	Consumer	Electronics	Bluetooth Speaker	3	17860	3048	0.1389	No	41 days	4	App Exclusive
LNKH-0104	2020-04-20 00:00:00	Eastern	Trincomalee	Online	Small Business	Electronics	Bluetooth Speaker	1	3480	3043	0.1009	No	41 days	4	None
LNKH-0106	2020-04-20 00:00:00	Central	Kandy	Online	Corporate	Home	Table Lamp	3	17812	7522	0.4445	No	41 days	4	None
LNKH-0208	2020-04-20 00:00:00	Eastern	Trincomalee	Online	Consumer	Beauty	Skincare Set	3	3514	1414	0.4000	No	1-2 days	5	Weekend Deal
LNKH-0104	2020-04-20 00:00:00	Central	Trincomalee	Online	Corporate	Electronics	Smart Watch	1	12410	1232	0.1232	No	1-2 days	4	New Year
LNKH-0103	2020-04-20 00:00:00	Central	Kandy	Online	Consumer	Beauty	Skincare Set	2	17832	6202	0.4040	No	41 days	100	App Exclusive
LNKH-0406	2020-04-20 00:00:00	Central	Nuwara Eliya	Online	Consumer	Sports	Dumbbell Set	2	10102	3102	0.3071	No	1-2 days	4	New Year
LNKH-0414	2020-04-20 00:00:00	Central	Malabe	Online	Consumer	Beauty	Hair Dryer	2	10888	4588	0.4118	No	41 days	3	None
LNKH-0415	2020-04-21 00:00:00	Eastern	Trincomalee	Online	Consumer	Perfume	Perfume	4	47460	10460	0.4399	No	1-2 days	5	None
LNKH-0419	2020-04-20 00:00:00	Eastern	Trincomalee	Online	Small Business	Home	Rice Cooker	2	24810	1410	0.1286	No	41 days	3	Weekend Deal
LNKH-0425	2020-04-20 00:00:00	Central	Nuwara Eliya	Online	Consumer	Beauty	Perfume	1	12322	2322	0.1883	No	1-2 days	4	App Exclusive
LNKH-0401	2020-04-20 00:00:00	Central	Malabe	Online	Small Business	Home	Air Fryer	1	23744	4244	0.1240	No	41 days	4	Weekend Deal
LNKH-0210	2020-04-20 00:00:00	Central	Malabe	Online	Consumer	Electronics	Smart Watch	1	12322	2344	0.1840	No	41 days	4	New Year

Figure 4 : Detail order-line table for the filtered selection above, that displays individual transactions.

5.0 Recommendations and Limitations

Recommended: Explore the return pipeline for the Electronics. An in-depth analysis of the product listings, product packaging and post-purchase support for the Electronics category will probably prove the biggest contributor to improving profit as a significant drop in the number of Electronics returns will have the greatest impact on overall margin (high revenue share and the highest return rate of 10.0% compared to platform average of 5.1%).

Recommendation 2 is to transfer what has been learnt in Western provinces to less-performing provinces. Western is already the top province in terms of revenue, so if Western is good in some areas of channel mix, delivery time, and promotion use – and bad in others – then it would

be possible to improve the performance of Western without spending any additional money, but by simply providing these successful methods to the less performing provinces – Uva and Eastern.

Limitations: This analysis contains data from 6 months (January-June, 2026) from one retailer's perspective only and therefore cannot be used to measure the effect of the season on other trading periods (such as the end of year). Due to the small number of customers that returned a rating, a rating-based analysis resulted in excluding the 42 rows containing no customer rating, which may not represent the true distribution of ratings for all customers. Lastly, since this is a coursework dataset it may not provide enough information to make a business decision for production (e.g., supplier costs, marketing expenses)

6.0 References

LankaMart Retail Transactions Dataset. (2026). CIT308 LankaMart Retail Transactions Dataset. Dataset provided for the CIT308 Data Visualization Mid-Semester Assignment.

Pandas Development Team. (2026). pandas documentation. Available at: <https://pandas.pydata.org/docs/>

Plotly. (2026). Plotly Python documentation. Available at: <https://plotly.com/python/>

Streamlit. (2026). Streamlit documentation. Available at: <https://docs.streamlit.io/>

Python Software Foundation. (2026). Python 3 documentation. Available at: <https://docs.python.org/3/>

SLTC Research University. (2026). CIT308 Data Visualization – Mid-Semester Assignment Specification. Semester 05, Bachelor of Applied Information Technology.

The dashboard has been created with open-source Python 3.11, Pandas, Plotly Express and Streamlit. The visualisations created were chosen based on the module assignment and the principles of dashboard design. A range of visualisations were chosen (KPI cards, time-series charts, categorical comparison charts, geographical comparisons and relationship visualisations) based on the requirements and the principles of dashboard design outlined in the module assignment.

7.0 AI and External Assistance Declaration

7.1 AI Assistance Declaration

The use of Generative Artificial Intelligence (AI) tools was used as a supportive tool in the development of this coursework. AI tools were most frequently employed to enhance the report's structure and clarity, review Python code, recommend appropriate data-visualization methods, establish potential data-quality checks, and refine the written explanations.

The analysis and interpretation of the Lanka Mart dataset, the implementation of the dashboard, the visualization selection and the final conclusion were reviewed and adapted for this assignment. Content generated by AI wasn't considered correct by default and the final content was reviewed against the data and assignment specifications.

7.2 External Assistance Declaration

The dashboard was created by referencing technical documentation that was open and publicly available from the Python, Pandas, Plotly and Streamlit communities. These resources were used to learn about the syntax of the programming language, the functions to process data, and the visualisation features.

No external person actually completed the analysis, programming, development of the dashboard and report writing. No template or pre-built project from outside the system was copied. This final dashboard and report is created just for the CIT308 Data Visualization Mid-Semester Evaluation.